Focus position and angle

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1 Movement of focus

When a sample is placed in the converging part of the beam it will shift the location of the focus which in term will change the measured electric field at the detector. First

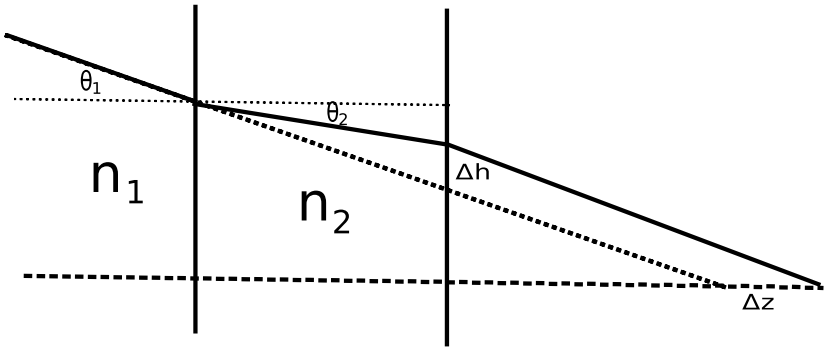

Figure 1: Illustration of the problem

to find Δh we subtract the distance from where the beams exit relative to where they entered the medium.

$$\Delta h = D\left(\tan\theta_1 - \tan\theta_2\right) \tag{1}$$

Now to get Δz

$$\Delta z = \frac{\Delta h}{\tan\theta_1} = D\frac{\tan\theta_1 - \tan\theta_2}{\tan\theta_1} \tag{2}$$

Following Snells law $n_1 \sin\theta_1 = n_2 \sin\theta_2$

$$\Delta z = D\frac{\tan\theta_1 - \tan\left[\sin^{-1}\left(\frac{n_1}{n_2}\sin\theta_1\right)\right]}{\tan\theta_1} \tag{3}$$

As n_1, n_2, θ_1 and D are all positive this goes to

$$\Delta z = D\left(1 - \frac{n_1\cos\theta_1}{\sqrt{n_2^2 - n_1^2\sin^2\theta}}\right) \tag{4}$$

2 Gaussian focus

The wavelength of THz is rather large and comparable to the change in the focus so to calculate the beam width on the detector we must use the equations for a Gaussian beam. The beam width as a function from the focus is

$$\omega\left(z\right)=\omega_0\sqrt{1+\left(\frac{\Delta z}{z_R}\right)^2} \tag{5}$$

where z_R is the Rayleigh range. Which is related to the beam waist ω_0.

$$z_R=\frac{\pi\omega_0^2}{\lambda} \tag{6}$$

We extract the full divergence angle of the beam so we can use this to get ω_0

$$\omega_0=\frac{2\lambda}{\pi\theta_1} \tag{7}$$

Putting this altogether we get

$$\omega\left(z\right)=\frac{2\lambda}{\pi\theta_1}\sqrt{1+\left(\frac{\Delta z\pi\theta_1^2}{4\lambda}\right)^2} \tag{8}$$

Its more useful if we put it in terms of frequency rather than wavelength

$$\omega\left(z\right)=\frac{2c}{f\pi\theta_1}\sqrt{1+\left(\frac{\Delta z f\pi\theta_1^2}{4c}\right)^2} \tag{9}$$

We measure the electric filed directly so if the spot at a given frequency the E field will reduce by the inverse. The magnitude of the ratio of transfer functions gives us the percentage change in electric field due to the change in focus, approximations and other unknowns. To get how much the E filed will reduce due to a change in spot size we can divide the beam waist at the focus with the beam width due to the change of focus position.

$$\frac{\omega_0}{\omega_\theta}=\frac{1}{\sqrt{1+\left(\frac{\Delta z f\pi\theta_1^2}{4c}\right)^2}} \tag{10}$$

3 Testing with teflon

The teflon sample we have is very thick, $D=21$ mm. Thus the focus moves quite a bit. Figure 2 is a time scan of the THz electric field with the sample in the collimated and the focus geometry. There is a clear reduction in signal for the focus geometry which is probably due to the focus of the THz beam moving. If the only thing the affects the

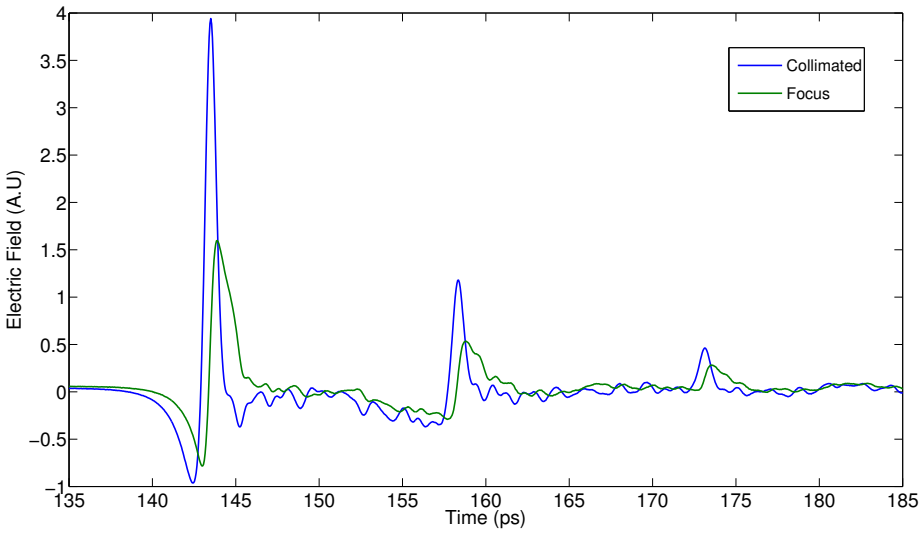

Figure 2: Time scan of E field with the PTFE sample in the collimated and converging beam. There is a clear reduction in the amplitude of the E field when in the focus geometry

amplitude of the E field is the movement of the focus then the amplitude of the ratio of the transfer functions T_{rat} should be equal to the change in E field given by equation 10.

$$|T_{rat}| = \left|\frac{T_{focus}}{T_{collimated}}\right| = \frac{1}{\sqrt{1+\left(\frac{\Delta z f \pi \theta_1^2}{4c}\right)^2}} \tag{11}$$

In figure 3 i have plotted both the amplitude of the transfer function and equation 10 for the extracted angles and extracted refractive index. This result gives the same slope bellow 0.5 THz but with an offset, maybe attenuation from absorption? After that the amplitude of T_{rat} goes back up which is not predicted.

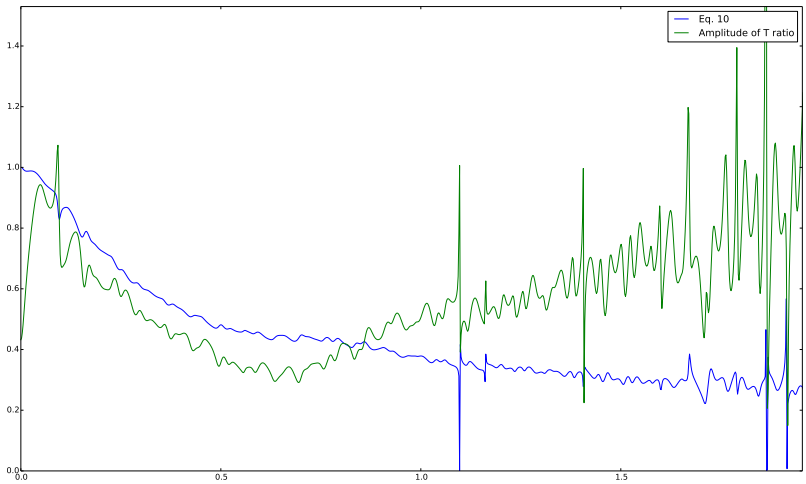

Figure 3: Amplitude of the transfer function and equation 10 plotted together

4 Fresnel equations

Another thing not taken into account is how the transmission depends on angle. Our emitter and detector are polarised and thus the beam we measure is p polarised to the surface of the sample. The Fresnell equation for transmission is

$$T_p = \frac{2n_1 \cos\theta_1}{n_1 \cos\theta_2 + n_2 \cos\theta_1} \tag{12}$$

The transmission through the sample is $T_{12}T_{21}$ To get how much this has an affect on T_{rat} we can divide the transmission though the sample at an angle and the transmission in the collimated case. Figure 4 is a plot of the ratio of transmission As you can see this does not do much. I included it anyway and plotted figure 5 which is a repeat of figure 3 but including the transmission coefficients.

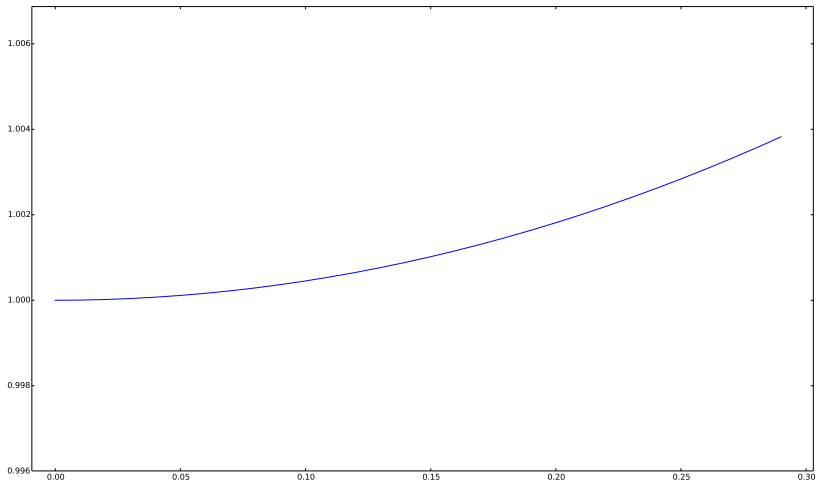

Figure 4: Ratio of transmission

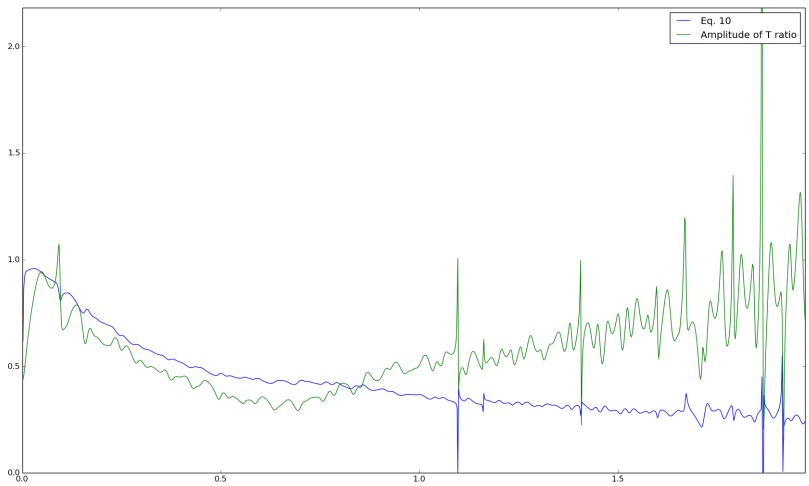

Figure 5: Amplitude of the transfer function and equation 10 plotted together but also including the transmission coefficients

5 To be continued

We still have not solved why the amplitude goes back up. I have not at any point taken into account that we should be integrating for a Gaussian beam which might solve the

problem. Another thing that is not taken into account is the hyper hemispherical Si lens on the detector. In figure 6 i have plotted the width of the THz pulse using equation 9 using the teflon data The Si lens has a diameter of 1 cm so this may also not be the problem.

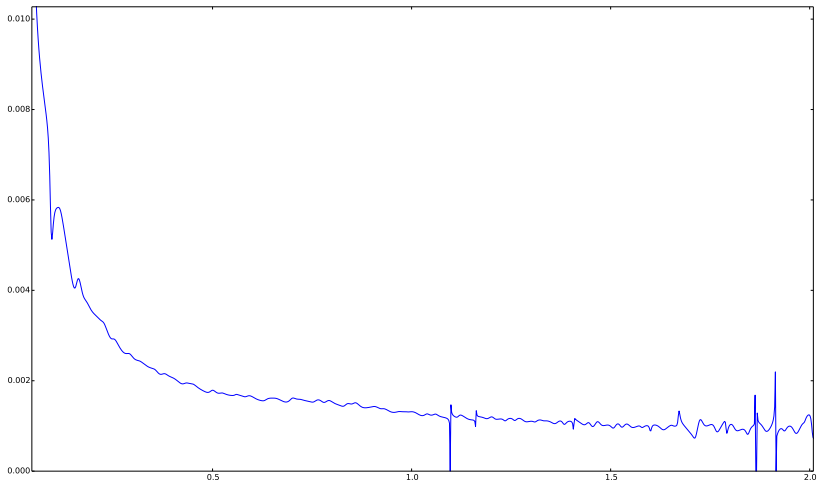

Figure 6: Beam width (radius)